

Def. Given two tangent vectors  $V_p$  and  $W_p$  at  $p \in \mathbb{R}^3$ , define

$$V_p \times W_p \stackrel{\text{def}}{=} \begin{bmatrix} 1 & 0 & 0 \\ V_1 & V_2 & V_3 \\ W_1 & W_2 & W_3 \end{bmatrix} \cdot (e_1)_p + \begin{bmatrix} 0 & 1 & 0 \\ V_1 & V_2 & V_3 \\ W_1 & W_2 & W_3 \end{bmatrix} \cdot (e_2)_p + \begin{bmatrix} 0 & 0 & 1 \\ V_1 & V_2 & V_3 \\ W_1 & W_2 & W_3 \end{bmatrix} \cdot (e_3)_p.$$

It is called the Cross Product of  $V_p$  and  $W_p$ .

Prop.  $V_p \times W_p = 0$  if and only if  $V_p$  and  $W_p$  are linearly dependent.

Proof. Suppose that  $V_p \times W_p = 0$ . Then, for any  $(u_1, u_2, u_3) \in \mathbb{R}^3$ ,

$$\det \begin{bmatrix} u_1 & u_2 & u_3 \\ V_1 & V_2 & V_3 \\ W_1 & W_2 & W_3 \end{bmatrix} = 0. \quad \text{So, taking } (u_1, u_2, u_3) \in \mathbb{R}^3 \setminus \text{Span}\{V, W\}, \text{ we obtain the result.}$$

(If  $V$  and  $W$  are linearly independent, then the determinant cannot be 0.)

Conversely, if  $V_p$  and  $W_p$  are linearly dependent, then regardless of the first row,  $\det = 0$ .

Lemma. Given  $V_p$  and  $W_p$  in  $T_p(\mathbb{R}^3)$ ,  $V_p \times W_p$  is orthogonal to both  $V_p$  and  $W_p$ ,

and  $\|V_p \times W_p\|^2 = (V_p \cdot V_p) \cdot (W_p \cdot W_p) - (V_p \cdot W_p)^2$ .

In other say,  $\|V \times W\| = \|V\| \cdot \|W\| \cdot \sin \theta$

Proof. By the linearity of the determinant,

$$V \cdot (V \times W) = \begin{vmatrix} V_1 & V_2 & V_3 \\ V_1 & V_2 & V_3 \\ W_1 & W_2 & W_3 \end{vmatrix} = 0 \quad \text{and} \quad W \cdot (V \times W) = \begin{vmatrix} W_1 & W_2 & W_3 \\ V_1 & V_2 & V_3 \\ W_1 & W_2 & W_3 \end{vmatrix} = 0.$$

Therefore, the orthogonality is proved. And, to show the second assertion,

$$(V \cdot V) \cdot (W \cdot W) - (V \cdot W)^2 = (V_1^2 + V_2^2 + V_3^2)(W_1^2 + W_2^2 + W_3^2) - (V_1 W_1 + V_2 W_2 + V_3 W_3)^2$$

$$= \sum_{1 \leq i, j \leq 3} V_i^2 \cdot W_j^2 - \left( \sum_{1 \leq i \leq j \leq 3} V_i W_i V_j W_j \right)$$

$$= \sum_{1 \leq i, j \leq 3} V_i^2 \cdot W_j^2 - \left( \sum_{i=1}^3 V_i^2 W_i^2 + 2 \cdot \sum_{1 \leq i < j \leq 3} V_i W_i V_j W_j \right)$$

$$= \sum_{1 \leq i < j \leq 3} V_i^2 \cdot W_j^2 - 2 \cdot \sum_{1 \leq i < j \leq 3} V_i W_i V_j W_j$$

Meanwhile,  $\|V \times W\|^2 = (V_2 W_3 - V_3 W_2)^2 + (V_1 W_3 - V_3 W_1)^2 + (V_1 W_2 - V_2 W_1)^2$

$$\begin{aligned} &= V_2^2 W_3^2 + V_3^2 W_2^2 - 2 V_2 W_2 V_3 W_3 \\ &\quad + V_1^2 W_3^2 + V_3^2 W_1^2 - 2 V_1 W_1 V_3 W_3 \\ &\quad + V_1^2 W_2^2 + V_2^2 W_1^2 - 2 V_1 W_1 V_2 W_2 \end{aligned}$$

These are same.  $\square$

